ForgeX

ForgeX Gen0 — Three-Phase Inverter Hardware Architecture & Design

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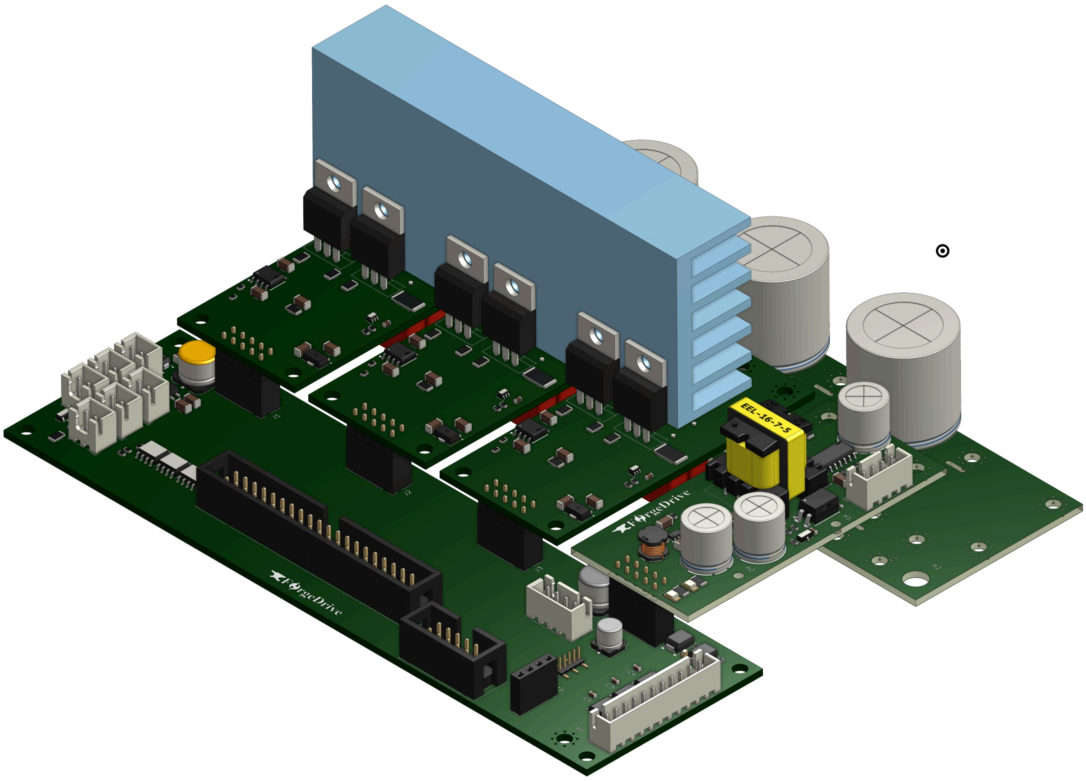
1. Introduction



1.1 Purpose

This document describes the architecture, design, implementation, and validation of the ForgeX Gen0 three-phase inverter platform.

It documents the hardware architecture used to realize the first ForgeX-generation power-conversion system, including its modular partitioning, electrical interfaces, power stage, control and measurement infrastructure, grounding strategy, PCB implementation, mechanical integration, and protection mechanisms.

The purpose of this document is to provide a technical reference for understanding how the ForgeX architectural principles are realized in the Gen0 implementation, as well as to record the engineering decisions, constraints, and trade-offs that shaped the design.

This document is intended to support continued development, maintenance, debugging, reproduction, and future evolution of the hardware.

1.2 Gen0 Scope

ForgeX Gen0 implements a modular three-phase voltage-source inverter intended primarily for motor-control and variable-speed drive applications.

The system is composed of multiple functional hardware modules, including the processing and development hardware, low-voltage signal and control infrastructure, power-bus interconnection, auxiliary power supply, and localized half-bridge power stages.

The Gen0 implementation provides the hardware infrastructure required for three-phase power conversion, including:

- Three-phase power switching
- High-voltage DC-bus distribution
- Localized gate-drive and power-stage circuitry
- Phase-current measurement
- DC-bus voltage and other system measurements
- PWM control interfaces
- Encoder and position-sensing interfaces
- Fault and protection signaling
- Auxiliary low-voltage power
- External communication and expansion interfaces

Gen0 is primarily intended as a development and validation platform. Its architecture is nevertheless designed to support practical deployment and to provide a foundation for subsequent ForgeX generations.

The scope of this document is limited to the Gen0 hardware implementation. Firmware, control algorithms, application software, and motor-control theory are discussed only where they directly affect the hardware architecture or its interfaces.

1.3 Design Objectives

The ForgeX Gen0 hardware was developed with the following objectives:

- **Modularity** — Separate major engineering functions into self-contained hardware modules with defined interfaces.
- **Electrical Locality** — Keep high-current paths, high-di/dt switching loops, gate-drive paths, local decoupling, and other electrically sensitive structures physically localized.
- **Maintainability** — Keep modules accessible for inspection, probing, troubleshooting, modification, and replacement.
- **R&D Accessibility** — Provide practical access to measurement points, interfaces, signals, and hardware during development and validation.
- **Scalability** — Establish an architecture that can be extended to different power levels, configurations, and power-conversion topologies without requiring a completely independent hardware architecture.
- **Interface Stability** — Establish clearly defined electrical, mechanical, and functional interfaces between modules.
- **Performance** — Achieve the electrical performance required of a practical power-conversion system without allowing modularity to unnecessarily degrade switching, measurement, thermal, or power distribution characteristics.
- **Incremental Development** — Allow individual modules to be developed, tested, characterized, and revised independently.
- **Practical Deployment** — Maintain an architectural path from experimental hardware toward a maintainable and field-deployable system.

These objectives are treated as engineering requirements and trade-offs rather than independent features. Where objectives conflict, electrical performance, safety, reliability, and well-defined architectural boundaries take precedence over modularity for its own sake.

1.4 Relationship to the ForgeX Family

ForgeX Gen0 is the first hardware implementation of the ForgeX family architecture.

It serves both as a functional three-phase inverter platform and as a reference implementation through which the architectural principles defined at the ForgeX family level are established and evaluated.

The Gen0 implementation realizes the ForgeX modular architecture through a set of functional hardware modules, including the HBM, LVP, PMB, PSU, and processing/development hardware.

The specific implementation of these modules, including component selection, circuit topology, PCB layout, mechanical construction, and interface details, is specific to Gen0 and is therefore documented within this generation-level document.

Gen0 interfaces are intended to establish an initial compatibility baseline for the ForgeX ecosystem. Future generations may preserve, modify, replace, or extend these interfaces according to the compatibility principles defined by the ForgeX family architecture.

Gen0 should therefore be regarded as both a complete inverter implementation and the first architectural reference point for future ForgeX generations.

2. Gen0 System Architecture

2.1 System Overview

ForgeX Gen0 modular three-phase voltage-source inverter designed primarily for motor-control and variable-speed drive applications.

The system separates the processing and control domain from the power-conversion hardware through a set of dedicated hardware modules. The primary power path is established through the PMB, while individual HBM modules provide the three localized half-bridge switching stages.

The LVP provides the low-voltage signal, measurement, sensing, and control infrastructure required to interface the processing domain with the power-stage hardware. The PSU provides the auxiliary low-voltage power required by the control, signal, and power-stage subsystems.

The resulting architecture separates the major electrical concerns while maintaining short and localized high-current and high-di/dt paths within the power-stage modules.

At the system level, the principal functional domains are:

- **Processing** — real-time control, PWM generation, communications, and system-level coordination.
- **Low-voltage control and sensing** — signal conditioning, measurement, encoder interfaces, protection, and control interfaces.
- **Power distribution** — DC-bus distribution and interconnection between power-stage modules.
- **Power conversion** — three independent half-bridge stages forming the three-phase inverter.
- **Auxiliary power** — generation and distribution of the required low-voltage supply rails.

The modular arrangement allows the Gen0 inverter to be assembled from independently developed subsystems while maintaining defined electrical, mechanical, and functional interfaces between them.

2.2 Functional Block Diagram

The following diagram illustrates the principal functional relationships between the ForgeX Gen0 hardware modules and the external system.

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The diagram distinguishes the primary power path from control, measurement, and communication paths and represents the architectural boundaries between the major hardware subsystems.

The PMB establishes the common DC-bus infrastructure and provides the physical interconnection through which the individual power-stage modules are combined into the three-phase inverter.

The three HBM modules constitute the three switching legs of the inverter. They receive their control signals from the low-voltage control domain (LVP) while obtaining their primary power through the PMB.

The PSU supplies the auxiliary low-voltage rails required by the ForgeX modules and associated control hardware.

The LVP provides the principal interface between the processing domain and the power-stage hardware including measurement, encoder, protection, and control-related interfaces.

3. Gen0 3-Phase Inverter Hardware Modules

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The ForgeX Gen0 three-phase inverter is implemented as a collection of functional hardware modules, each responsible for a defined portion of the overall system.

The module boundaries follow the architectural principles established by the ForgeX family: power conversion, power distribution, low-voltage control, auxiliary power, and processing are separated into localized and independently replaceable subsystems.

The following modules constitute the primary Gen0 hardware platform. Detailed circuit design, PCB implementation, interfaces, constraints, and validation results for each module are documented in the corresponding sections.

3.1 LVP_G0S4

Low-Voltage Plane — Gen0, 4 Slots

The LVP provides the primary low-voltage control and signal interconnection infrastructure for the Gen0 system.

It interfaces the processing domain with the power-stage modules and provides infrastructure for measurement, sensing, protection, encoder interfaces, and other low-voltage signals.

The `S4` designation indicates the four-slot Gen0 implementation, providing the physical interfaces required for the three HBM modules and the associated auxiliary/power module connections.

The LVP is intentionally kept separate from the high-current power conversion hardware, allowing the low-voltage control and sensing infrastructure to evolve independently of the power stage.

3.2 PMB_G0S4VH4

Power MotherBus — Gen0, 4 Slots, 400 V Class

The PMB provides the primary high-voltage power distribution infrastructure of the Gen0 inverter.

The `S4` designation indicates four power-module interfaces, while `VH4` identifies the 400 V-class voltage rating of the implementation.

The PMB distributes the DC bus to the three HBM modules and the PSU while providing a centralized and mechanically organized power interconnection structure.

The PMB is deliberately kept functionally simple in its fundamental role: it provides the infrastructure through which the individual power modules are combined into a complete converter. Where required, additional functionality such as bus measurement, protection, or precharge infrastructure can be introduced in future implementations without changing the fundamental role of the PMB.

3.3 HBM_G0VH4C5

Half-Bridge Module — Gen0, 400 V Class, 5 A

The HBM is the localized power-switching module of the ForgeX Gen0 inverter.

Each HBM implements one half-bridge of the three-phase inverter and contains the associated power switching devices, gate-drive circuitry, local measurement, protection, and supporting circuitry required to operate the switching stage.

Three HBM modules are combined through the PMB to form the three switching legs of the three-phase inverter.

The `VH4` designation identifies the 400 V-class implementation, while `C5` identifies the 5 A current class.

The HBM is designed around electrical locality: high-current paths, high-di/dt switching loops, gate-drive paths, current Sensing, module output voltae sensing, and local decoupling are contained within the module rather than distributed across the wider system.

3.4 PSU_G0P20N

Power Supply Module — Gen0, 20 W

The PSU provides the auxiliary low-voltage power required by the ForgeX Gen0 system.

The `P20` designation identifies the nominal 20 W power class of the module. The `N` suffix identifies an implementation with a dedicated interface exposing the high-voltage DC-bus ground and a non-isolated auxiliary low-voltage supply.

This interface allows the auxiliary supply reference to be connected to the system grounding topology at a defined star point. In non-galvanically-isolated LVP implementations, this provides a controlled means of establishing the required common reference between the auxiliary low-voltage domain and the DC-bus ground.

The PSU is treated as an independent subsystem so that auxiliary power generation and regulation can be developed, tested, and revised without requiring corresponding changes to the primary inverter power-stage hardware.

The module provides the required supply rails to the low-voltage, processing, and power-stage subsystems through the defined Gen0 interfaces.

3.5 BlackDev_G0Basic

Processing / Development Module — STM32F401/F411

`BlackDev_G0Basic` is the primary processing and development module used with the ForgeX Gen0 platform.

Rather than implementing a complete processing board, the `BlackDev_G0Basic` is primarily a carrier and interface shield for STM32F401/F411-based Black Pill development boards. It provides the electrical and mechanical infrastructure required to integrate the Black Pill into the ForgeX system while exposing the required ForgeX interfaces to the remainder of the hardware.

The underlying Black Pill provides the primary microcontroller and its associated basic support circuitry. `BlackDev_G0Basic` provides the additional interconnection, signal routing, protection, connectors, and development infrastructure required to integrate the processor with the ForgeX Gen0 architecture.

This approach deliberately avoids unnecessarily reproducing a complete microcontroller development platform within ForgeX. It allows the processing element to remain inexpensive, widely available, easily replaceable, and convenient to modify or experiment with during R&D.

The module provides the computational platform used for real-time control, PWM generation, measurement processing, communication, and system-level coordination.

The processing module communicates with the remaining ForgeX hardware through defined low-voltage interfaces rather than requiring the microcontroller itself to be physically integrated into the power electronics.

The resulting separation also allows the underlying Black Pill to be replaced or upgraded independently of the primary ForgeX power hardware, provided the relevant processing and interface requirements are maintained.

Gen0 Module Overview

Module	Role	Primary Rating / Characteristic
`LVP_G0S4`	Low-voltage control and signal infrastructure	4-slot
`PMB_G0S4VH4`	High-voltage power distribution	4-slot, 400 V class
`HBM_G0VH4C5`	Half-bridge power stage	400 V class, 5 A
`PSU_G0P20N`	Auxiliary power	20 W
`BlackDev_G0Basic`	Processing / development	STM32F401/F411

These modules form the baseline Gen0 hardware ecosystem. The following sections describe their individual electrical architectures, interfaces, PCB implementation, design constraints, and validation.

4. System Level Layout and Design

4.1 Power Distribution Network (PDN) and Grounding Scheme

The ForgeX Gen0 system employs a bus-like power distribution structure, with the primary DC-bus interconnection centralized at the PMB.

The power modules are connected directly to the common DC-bus structure, minimizing the length and inductance of the high-current power paths. This approach reduces parasitic DC-bus inductance and consequently reduces the voltage developed across the distribution network during rapid current transients:

$$V_L = L\frac{di}{dt}$$

Minimizing this parasitic inductance is particularly important for the high-di/dt switching currents generated by the HBM power stages. The resulting reduction in parasitic voltage excursion helps limit switching-node overshoot, ringing, and associated electromagnetic interference.

The grounding architecture follows the same principle. Power and low-voltage reference connections are organized around a defined system grounding point at the power entry, with the relevant module returns arranged to minimize unwanted circulating currents and shared impedance.

This star-oriented grounding strategy reduces common impedance between high-current power paths and sensitive control and measurement references. Consequently, voltage disturbances produced by switching currents are less likely to appear as ground bounce at sensitive interfaces.

Together, the low-inductance power distribution and controlled grounding topology provide a defined current-return structure for the system and help reduce parasitic coupling, common-mode disturbances, and electromagnetic interference.

5. Module Level Overview and External Interfaces

5.1 PMB_G0S4VH4 Brief

Revision: revA
Design Files
Schematic
Technical Reference (*Work in Progress* 🛠)

The PMB employs a biplanar laminated DC-bus structure in place of the originally considered coplanar bus arrangement. The change was primarily motivated by the lower parasitic inductance of the biplanar structure, with the additional objective of reducing high-frequency voltage disturbances, electromagnetic interference, and inter-module ground potential variation.

The resulting bus geometry provides a relatively short and tightly coupled current loop. For the farthest module, corresponding to a maximum bus path length of approximately 15 cm, width of 10mm and spacing of 1.6mm, the analytical worst-case loop inductance is estimated at approximately 30 nH for the implemented geometry.
This estimate is based on the biplanar bus geometry and the corresponding calculation provided by the EMI Software biplanar bus calculator:

EMI Software — Biplanar Bus Inductance Calculator

Distributed DC-link capacitors are placed in close proximity to the individual power modules. This reduces the effective high-frequency current-loop area and minimizes the amount of switching current that must flow through the main DC-bus structure. Local placement also reduces the effective parasitic ESL seen by each HBM during switching transients.

For the farthest module, the estimated equivalent series resistance (ESR) of the DC-link path from the module to the primary DC input terminal is below 10 mOhm.

The PDN was designed with explicit limits on both resistive and inductive disturbances. Under the nominal 1 kW, 400 V operating condition, the expected DC common-mode voltage variation between modules is targeted at less than 50 mV, while inductively induced ground-bounce voltage is targeted at less than 200 mV.

These limits were used as design targets for the PMB geometry, distributed DC-link capacitance, grounding topology, and physical placement of the power modules.

The connector pad matrix serves a dual purpose as both the high-current electrical termination for the module and its primary mechanical fixation interface. M3 copper or brass spacers are used as the mechanical fastening elements and, where electrically connected, provide a low-impedance current path between the module and the PMB.

The fastening arrangement is designed for firm mechanical clamping to maintain low and stable contact resistance while providing sufficient mechanical rigidity for the assembled power structure.

5.2 LVP_G0S4 Brief

Revision: revA
Design Files
Schematic
Technical Reference (*Work in Progress* ⚒)

The `LVP_G0S4` implements the primary low-voltage control, signal conditioning, sensing, and expansion infrastructure of the ForgeX Gen0 platform.

The current revision exposes the following primary feature set.

**Encoder Interfaces

The LVP provides native physical-layer support for several encoder interface types, allowing the same hardware platform to accommodate different feedback technologies without requiring changes to the primary processing hardware.

Supported interfaces include:

- SSI
- BiSS-C
- A/B/Z quadrature encoders
- Differential Hall-effect encoder signals

The encoder interfaces are routed through the LVP's low-voltage signal infrastructure and are presented to the processing domain through the defined system interfaces.

Processing and Communication (PC40G0) Interface

The `LVP_G0S4` PC40G0 Interface exposes a packed 40-pin system connector carrying the primary low-voltage signals exchanged between the processing domain, power-stage modules, sensor interfaces, and communication hardware.

The connector consolidates the principal digital and analog signals required for system integration while maintaining a defined interface between the LVP and the remainder of the ForgeX hardware.

Expansion Interfaces

The LVP provides multiple expansion interfaces, including:

- Expansion SPI
- Expansion UART

These interfaces are intended to support distributed functionality within the ForgeX architecture.

Rather than requiring every function to remain directly connected to the primary processor, future implementations may assign specialized responsibilities to additional devices or modules. Examples include dedicated encoder-interface devices, intelligent gate-driver controllers, distributed measurement devices, or other specialized peripherals.

This provides a path toward increasing system capability without requiring a corresponding increase in complexity on the primary processing module.

Temperature and Fan Interfaces

The LVP provides three channels of 100 kOhm NTC temperature-sensor signal conditioning for monitoring temperatures at multiple locations within the system.

Three fan interfaces are also provided. In the Gen0 implementation, the fan outputs are currently controlled through a common control scheme rather than independently controlled channels.

The interfaces provide the basic infrastructure required for thermal management while leaving room for more sophisticated fan-control strategies in future generations.

High-Speed Overcurrent Protection

The LVP implements three high-speed comparator channels for overcurrent protection.

The comparator paths are designed for a response time below approximately 150 ns and provide a hardware-level protection path independent of normal firmware execution.

The resulting fault signals are forwarded to the MCU timer break-input path (`~BKIN`), allowing the PWM hardware to rapidly disable the power stage following detection of an overcurrent condition.

This architecture provides a fast protection mechanism while keeping the protection decision outside the normal software control loop.

High-Speed Digital Layout

The LVP PCB layout is designed around the electrical requirements of its high-speed digital interfaces.

Digital interfaces are primarily routed toward the right-hand side of the board, maintaining a defined signal-flow direction and minimizing unnecessary crossings through unrelated circuit regions.

High-speed signal traces are routed over continuous reference planes where practical, avoiding interruptions in the return-current path and minimizing unnecessary loop area.

The layout was developed with a target of supporting approximately 10 MHz BiSS-C operation and 40 MHz SPI operation without significant signal degradation under the intended interconnect conditions.

These frequencies represent design targets rather than guaranteed maximum interface rates; final achievable performance depends on the connected devices, trace geometry, connector characteristics, termination, loading.

5.3 HBM_G0VH4C5 Brief

Revision: revA
Design Files
Schematic
Technical Reference

The `HBM_G0VH4C5` (Half-Bridge Module) implements one switching leg of the ForgeX Gen0 three-phase inverter.

The module integrates the power switching devices, gate-drive circuitry, current measurement, switch-node voltage measurement, and the supporting protection and control circuitry required to operate the half-bridge as a self-contained power-stage module.

The HBM is responsible for controlling the switching state of its power devices to perform the power modulation required by the inverter. The `Gen0 VH4C5` implementation uses a low-side shunt for output current measurement and provides switch-node voltage measurement for monitoring and control purposes.

A primary design objective of the HBM is to maintain **electrical locality** within the power stage. High-di/dt switching paths, gate-drive paths, local DC-link connections, and associated return paths are kept physically compact in order to minimize parasitic inductance, switching-loop area, and unwanted coupling between the power and control domains.

The module interfaces with the PMB through the defined high-current power-module interface and exposes a low-voltage control and measurement interface toward the LVP.

Three HBM modules are combined through the `PMB_G0S4VH4` to form the three switching legs of the ForgeX Gen0 three-phase inverter.

The HBM therefore establishes the primary boundary between the high-energy switching domain and the low-voltage control and measurement infrastructure of the ForgeX architecture.

5.4 PSU_G0P20N Brief

Revision: revA
Design Files
Schematic
Technical Documentation

The `PSU_G0P20N` provides the auxiliary low-voltage power required by the ForgeX Gen0 platform. The module is designed with galvanic isolation between the primary high-voltage input domain and its main low-voltage outputs.

For the current Gen0 implementation, the primary auxiliary outputs exposed by the module are **12 V and 5 V**, providing the supply rails required by the ForgeX control, processing, and supporting circuitry.

Although the auxiliary outputs are galvanically isolated by design, the PSU also exposes the high-voltage power-domain ground through a dedicated interface. This allows the isolation boundary to be selectively bridged where required by the system architecture.

In particular, this provides a controlled grounding option for non-galvanically-isolated LVP implementations, where the auxiliary low-voltage reference may be intentionally tied to the system power ground at the defined grounding point.

The PSU therefore supports both isolated and selectively non-isolated system configurations without requiring a fundamentally different auxiliary power module.

🛠UNDER DEVELOPMENT🛠

Sections Under are mainly still to be implemented

10. Validation and Bring-Up

10.1 Initial Bring-Up

10.2 Low-Voltage Testing

10.3 Gate-Drive Validation

10.4 PWM Validation

10.5 ADC / Measurement Validation

10.6 Switching Validation

10.7 Thermal Validation

10.8 Protection Validation

10.9 Motor Testing

11. Known Issues and Limitations

11.1 Known Hardware Issues

11.2 Performance Limitations

11.3 Documentation Gaps

11.4 Mechanical Assembly

13. Future Development

13.1 Gen0 Improvements

13.2 Potential Gen1 Changes

13.3 Interface Evolution

13.4 Performance Improvements

13.5 Additional Modules

14. Revision History